Building a Pose Map

1. Course Content

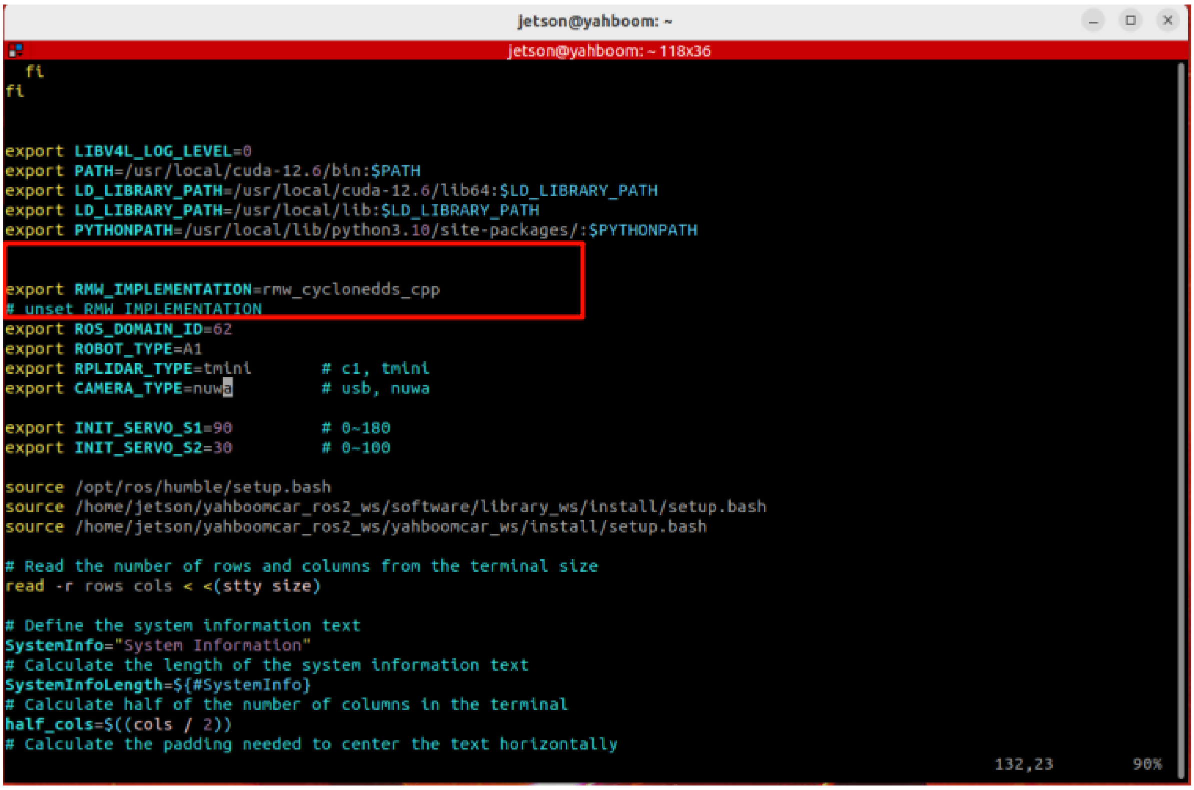
1. Building Raster Maps and Pose Maps using slam_toolbox

> [!NOTE]
>
> - At least one raster map is required before the road network .geojson file.
> - The pose map is used for fast relocalization during waypoint marking and navigation, improving the accuracy of global localization and thus the accuracy of waypoint marking.
> - This demonstration uses the **Orin board**.

2. Building a Raster Map

Open ~/.bashrc, uncomment this line, and change it as shown in the image. (After learning about road network planning, please restore this comment to avoid affecting previous examples.)

```
vim ~/.bashrc
:wq #Save and exit
source ~/.bashrc #Update environment
```



> [!TIP]

> - Before building the map, you can place some obstacles outside the roads on the map so that the actual position of the car can be referenced during navigation.

- Start slam_toolbox to build the map,

```
ros2 launch yahboomcar_nav map_cartographer_launch.py
```

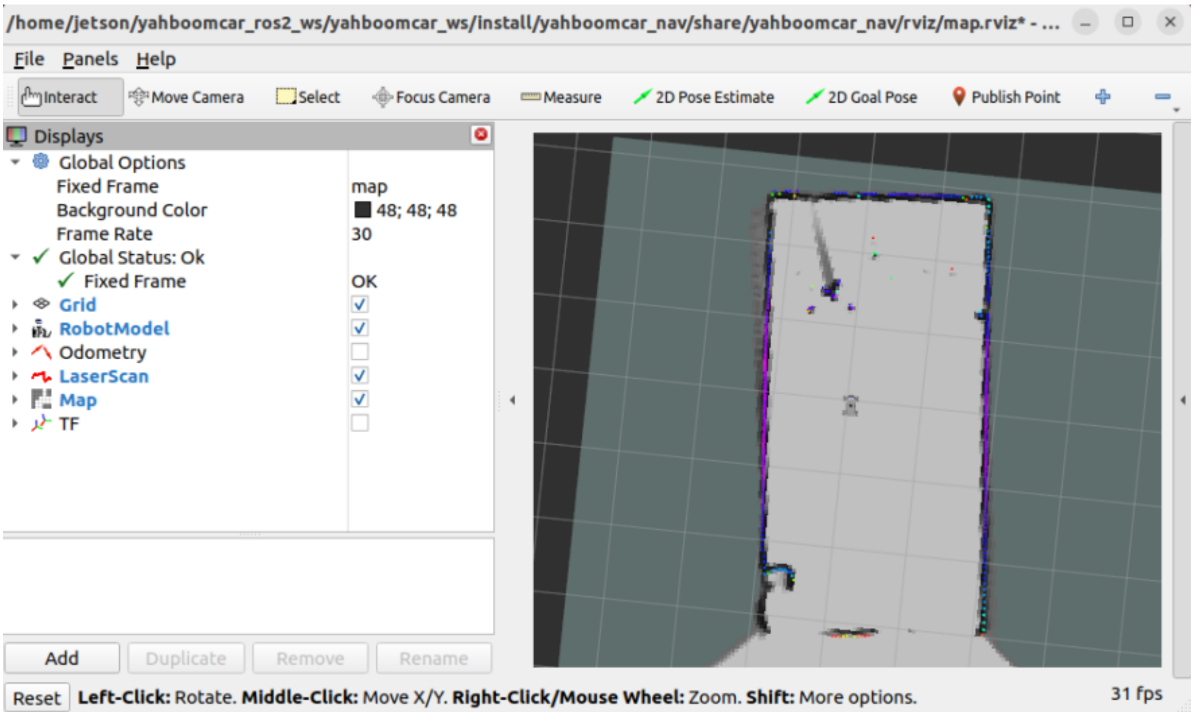
- Launch RViz visualization,

```
ros2 launch yahboomcar_nav display_map_launch.py
```

- Launch the keyboard control node (a gamepad can also be used),

```
ros2 run yahboomcar_ctrl yahboom_keyboard
```

- Control the car's movement to build the map

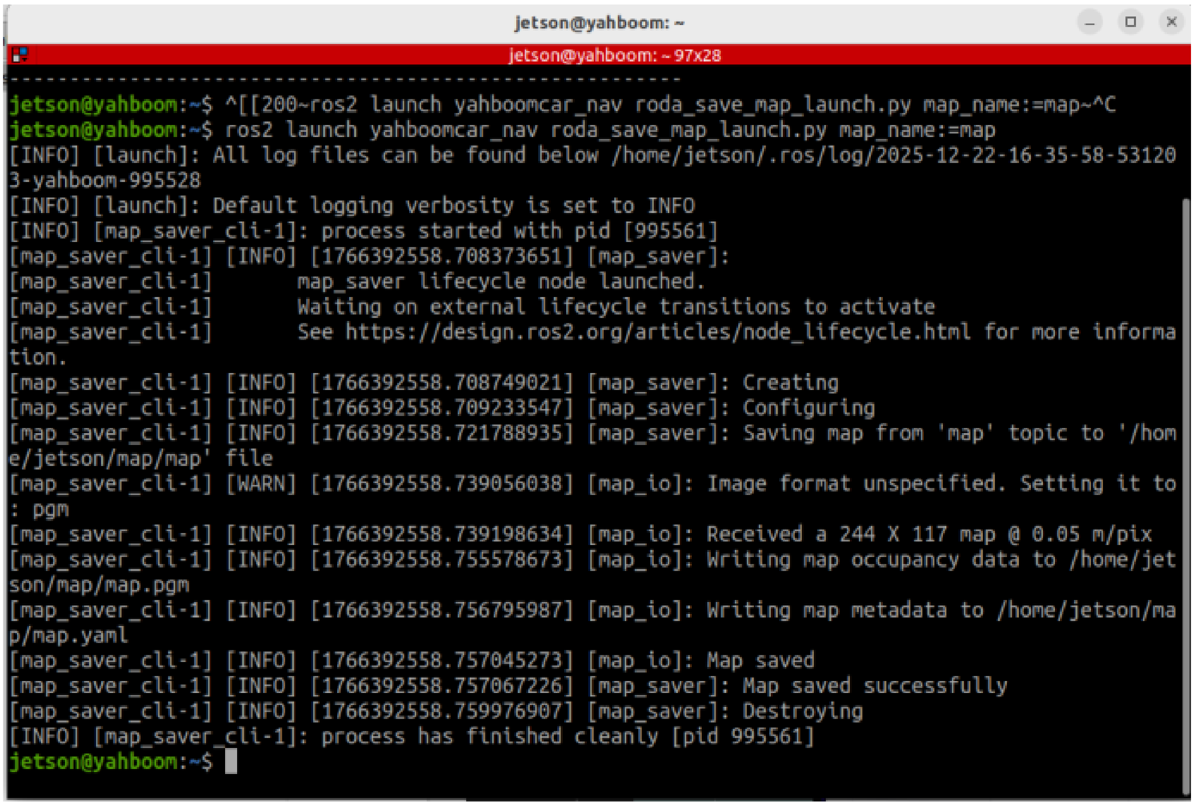


- Save the raster map

```
#ORIN
 ros2 launch yahboomcar_nav roda_save_map_launch.py map_name:=map
```

> [!TIP]
>
> - map_name specifies the **map name** for the saved raster map. If no startup parameter is specified, the default map name is `map`. All map files are saved in the `~/map` directory.

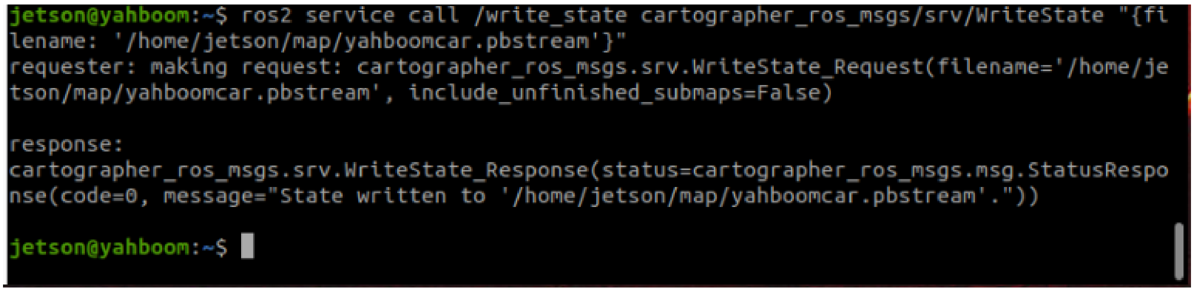
The terminal message `Map saved successfully` indicates that the map was saved successfully.



```
jetson@yahboom: ~
jetson@yahboom: ~ 97x28
------------------------------------------------------------------
jetson@yahboom:~$ ^[[200~ros2 launch yahboomcar_nav roda_save_map_launch.py map_name:=map~^C
jetson@yahboom:~$ ros2 launch yahboomcar_nav roda_save_map_launch.py map_name:=map
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2025-12-22-16-35-58-53120
3-yahboom-995528
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [map_saver_cli-1]: process started with pid [995561]
[map_saver_cli-1] [INFO] [1766392558.708373651] [map_saver]:
[map_saver_cli-1]       map_saver lifecycle node launched.
[map_saver_cli-1]       Waiting on external lifecycle transitions to activate
[map_saver_cli-1]       See https://design.ros2.org/articles/node_lifecycle.html for more informa
tion.
[map_saver_cli-1] [INFO] [1766392558.708749021] [map_saver]: Creating
[map_saver_cli-1] [INFO] [1766392558.709233547] [map_saver]: Configuring
[map_saver_cli-1] [INFO] [1766392558.721788935] [map_saver]: Saving map from 'map' topic to '/hom
e/jetson/map/map' file
[map_saver_cli-1] [WARN] [1766392558.739056038] [map_io]: Image format unspecified. Setting it to
: pgm
[map_saver_cli-1] [INFO] [1766392558.739198634] [map_io]: Received a 244 X 117 map @ 0.05 m/pix
[map_saver_cli-1] [INFO] [1766392558.755578673] [map_io]: Writing map occupancy data to /home/jet
son/map/map.pgm
[map_saver_cli-1] [INFO] [1766392558.756795987] [map_io]: Writing map metadata to /home/jetson/ma
p/map.yaml
[map_saver_cli-1] [INFO] [1766392558.757045273] [map_io]: Map saved
[map_saver_cli-1] [INFO] [1766392558.757067226] [map_saver]: Map saved successfully
[map_saver_cli-1] [INFO] [1766392558.759976907] [map_saver]: Destroying
[INFO] [map_saver_cli-1]: process has finished cleanly [pid 995561]
jetson@yahboom:~$
```

- Save the cartograph pose map

```
ros2 service call /write_state cartographer_ros_msgs/srv/WriteState "{filename:
'/home/jetson/map/map.pbstream'}"
```

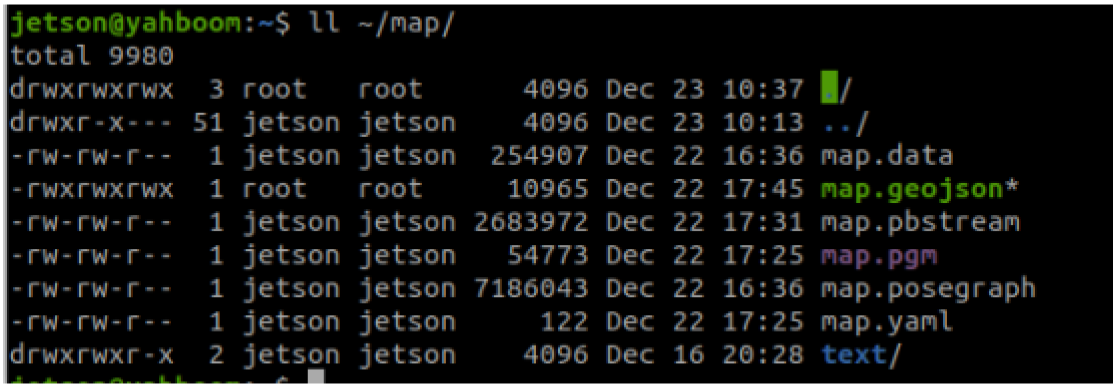


```
jetson@yahboom:~$ ros2 service call /write_state cartographer_ros_msgs/srv/WriteState "{fi
lename: '/home/jetson/map/yahboomcar.pbstream'}"
requester: making request: cartographer_ros_msgs.srv.WriteState_Request(filename='/home/je
tson/map/yahboomcar.pbstream', include_unfinished_submaps=False)

response:
cartographer_ros_msgs.srv.WriteState_Response(status=cartographer_ros_msgs.msg.StatusRespo
nse(code=0, message="State written to '/home/jetson/map/yahboomcar.pbstream'."))

jetson@yahboom:~$
```

- Map files are saved by default in the `~/map` path. You can also specify the map name using the `map_name` parameter when saving the map.



```
jetson@yahboom:~$ ll ~/map/
total 9980
drwxrwxrwx  3 root   root      4096 Dec 23 10:37 ./
drwxr-x--- 51 jetson jetson    4096 Dec 23 10:13 ../
-rw-rw-r--  1 jetson jetson  254907 Dec 22 16:36 map.data
-rwxrwxrwx  1 root   root     10965 Dec 22 17:45 map.geojson*
-rw-rw-r--  1 jetson jetson 2683972 Dec 22 17:31 map.pbstream
-rw-rw-r--  1 jetson jetson   54773 Dec 22 17:25 map.pgm
-rw-rw-r--  1 jetson jetson 7186043 Dec 22 16:36 map.posegraph
-rw-rw-r--  1 jetson jetson     122 Dec 22 17:25 map.yaml
drwxrwxr-x  2 jetson jetson    4096 Dec 16 20:28 text/
```